DSA4262 Sense-making Case Analysis: Health and Medicine

Project Phase: Overview

19-10-2026

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www.jglab.org

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Outline for today

- Upcoming Deadlines
- Project Phase: Overview
  - Task 2: Analyse, visualise, summarise genomic data
- Data Visualisation
- Hackathon (3/4) and team meetings

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Project Phase: Overview

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Project Phase: Overview

L5 (14.09): Team Project Introduction: Identification of m6A RNA modifications from direct RNA-Seq data

L6 (28.09): Machine Learning in Genomics

L7 (5.10): Intermediate Leaderboard submission + Hackathon (I)

-> First leaderboard submissions due **Oct 7th, 23.59pm**

L8 (12.10): Intermediate Leaderboard: Results + Best Practice + Hackathon (II)

L9 (19.10): Analysis and visualisation of genomics data + Hackathon (III)

L10 (26.10): Final Hackathon (IV): No questions will be answered anymore after this lecture

-> Final submission due **Oct 28th, 23.59pm**

-> teamwork summary due **Oct 29th, 23.59pm.**

L11 (02.11): Summary, Final Results and Awards Ceremony

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Example timeline

Intermediate leaderboard submission

this week

final submission

	Week 6	Week 7	Week 8	Week 9	Week 10	Status
Data parsing						Completed
First model						Completed
Final model						Completed
Code documentation + executable code						Completed
Analysis of SG-NEx samples						Ongoing
Interactive visualisation						Ongoing
Report						Ongoing

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Budget Overview

- All teams within budget
  - Min: 0$
  - Max: 224$
- Monitor your project budget
- Please remember to keep budget for the evaluation of other teams as well.

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Project Overview and Assessment Components

- **Report** (including interactive data visualisation) (50%)
- **Model performance/Leaderboard submission**: results will be evaluated using an unknown test dataset and compared against a random classifier and a baseline model (with ROC AUC and Precision-Recall AUC) (5%)
  - the ranking in comparison with other teams will not be used for grading
- **Documentation and Reproducibility**: code needs to be executable and results need to be reproducible (5%) - *this will be evaluated by other students*
- **Individual component:** (5%)
  - Each student is expected to attend in person and actively contribute to the project solution
  - Each student should have a specific contribution in report writing, model development, and implementation
  - Evaluation will be based on confidential peer reports

Please refer to the Team Project folder on Canvas for complete details and requirements

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Task 2: Prediction of m6A sites in all SG-NEx direct RNA-Seq data sets

- **Predict m6A RNA modifications** in all direct RNA-Seq data sets from the **SG-NEx data** using your method
- **Describe the results** and **compare them** across the different cell lines
- **Summarise** and **visualise** your observations
- Create an **interactive data visualisation platform** for other researchers to explore this data

The task is broadly defined on purpose:

- Demonstrate that you can interpret data and predictions within a specific domain/context
- Identify relevant questions
- Communicate results

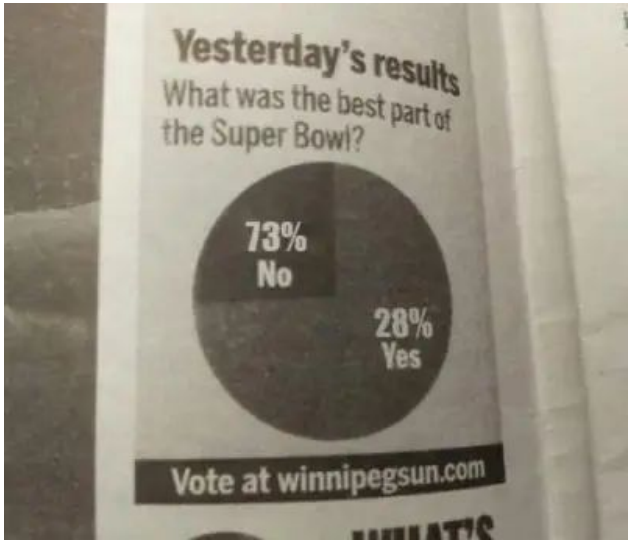
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Scientific Figure Design

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How to design figures

“Figures are powerful tools to effectively and efficiently convey complex information.”



Rolandi, Marco, Karen Cheng, and Sarah Pérez-Kriz. "A brief guide to designing effective figures for the scientific paper." Advanced Materials 23.38 (2011): 4343-4346.

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How to design scientific figures

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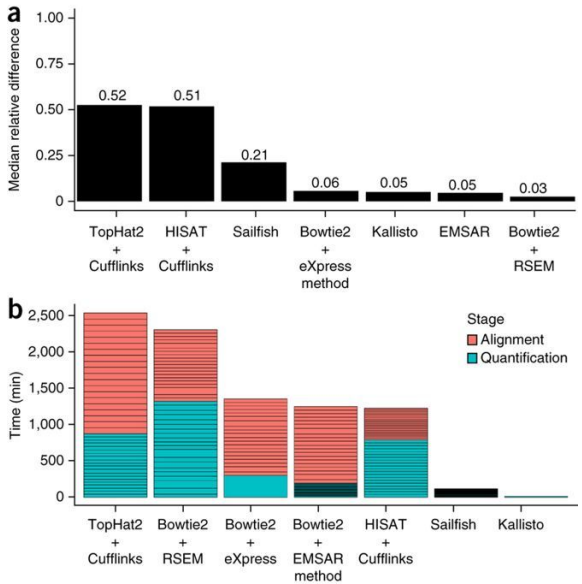
Bad figures are powerful tools to effectively and efficiently confuse your audience and undermine the trust in your work.

Today: 10 Simple rules for better figures

Rolandi, Marco, Karen Cheng, and Sarah Pérez-Kriz. "A brief guide to designing effective figures for the scientific paper." Advanced Materials 23.38 (2011): 4343-4346.

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What is good or bad in this figure?



(a) Accuracy of kallisto, Cufflinks, Sailfish, EMSAR, eXpress and RSEM on 20 RSEM simulations of 30 million 75-bp paired-end reads based on the abundances and error profile of GEUVADIS sample NA12716_7 (selected for its depth of sequencing). For each simulation, we report the accuracy as the median relative difference in the estimated read count of each transcript. Estimated counts were used rather than transcripts per million (TPM) because the latter is based on both the assignment of ambiguous reads and the estimation of effective lengths of transcripts, so a program might be penalized for having a differing notion of effective length despite accurately assigning reads. The values reported are means across the 20 simulations (the variance was too small to be visible in this plot). Relative difference is defined as the absolute difference between the estimated abundance and the ground truth divided by the average of the two.

(b) Total running time in minutes for processing the 20 simulated data sets of 30 million paired-end reads described in a. All processing was done using 20 cores, with programs being run with 20 threads when possible (Bowtie2, TopHat2, RSEM, Cufflinks) and 20 parallel processes otherwise (eXpress, kallisto). Each box represents one dataset. Since eXpress and kallisto process all datasets in parallel, the only quantification time shown is the maximum of all the quantifications.

What is good?

- Clear message
  - Kallisto is accurate and very fast
- Simple design
- Known plot type (barplot)

Criticism

- Misalignment
- Black: too strong?
- No variance (summary of summary)
- Data is highly processed
- Stacked: individual not visible
- (Parallel vs threads: unfair comparison, sum is also unfair to slower methods)

Bray, Nicolas L., et al. "Near-optimal probabilistic RNA-seq quantification." Nature Biotechnology 34.5 (2016): 525-527.

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(1) The purpose of a figure is to convey a message

“A figure is meant to express an idea or introduce some facts or a result that would be too long (or nearly impossible) to explain only with words[...].

In this context, it is important to clearly identify the role of the figure, i.e., **what is the underlying message and how can a figure best express this message?**

Once clearly identified, this message will be a strong guide for the design of the figure[...]. Only after identifying the message will it be worth the time to develop your figure, just as you would take the time to craft your words and sentences when writing an article only after deciding on the main points of the text. If your figure is able to convey a striking message at first glance, chances are increased that your article will draw more attention from the community.”

Quote from: Rougier, Nicolas P., Michael Droettboom, and Philip E. Bourne. "Ten simple rules for better figures." PLoS computational biology 10.9 (2014): e1003833.https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1003833

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(1) The purpose of a figure is to convey a message

1) Identify the message
2) Design the figure, minimise and remove anything that does not directly contribute to conveying the message
3) The message should be clear at first glance

Quote from: Rougier, Nicolas P., Michael Droettboom, and Philip E. Bourne. "Ten simple rules for better figures." PLoS computational biology 10.9 (2014): e1003833.https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1003833

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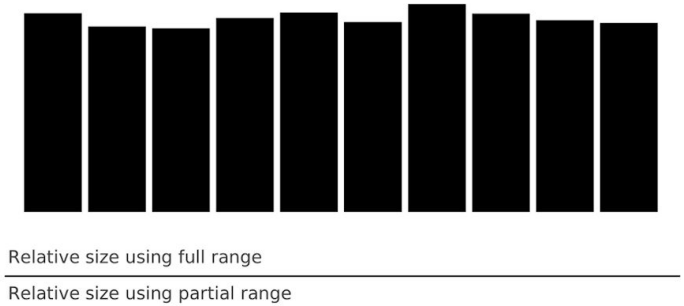
(2) Do Not Mislead the Reader



Rougier, Nicolas P., Michael Droettboom, and Philip E. Bourne. "Ten simple rules for better figures." PLoS computational biology 10.9 (2014): e1003833.https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1003833

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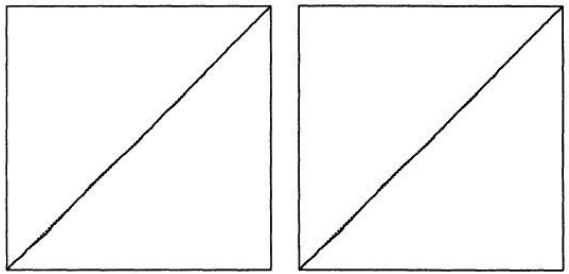


- A figure summarises data to convey a message and guide interpretation of the data
- The message should be supported by the data
- Visualisations can easily be used to suggest data interpretation that is not sufficiently supported by data

-> Do not mislead the reader

Rougier, Nicolas P., Michael Droettboom, and Philip E. Bourne. "Ten simple rules for better figures." PLoS computational biology 10.9 (2014): e1003833.https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1003833

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Mykland P, Tierney L, Yu B (1995) Regeneration in Markov chain samplers. Journal of the American Statistical Association 90:233-241, Figure 1, https://www.biostat.wisc.edu/~kbroman/topten_worstgraphs/

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(3) Describe your figure

Each figure should be described by:

1. Figure labels
    a. x-axis labels
    b. y-axis labels
    c. scale
    d. Legend (colour/line)
2. A figure caption describes the figure details

- The figure should be interpretable when looking at it (without reading the caption or text).
- Figure labels need to be of readable size

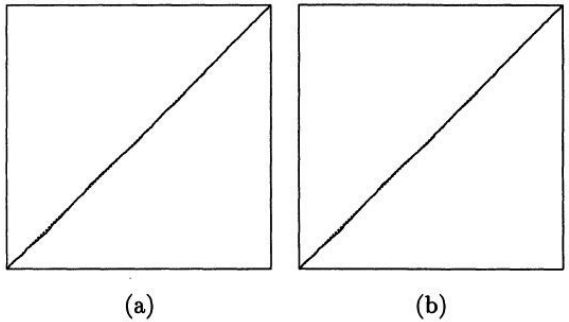
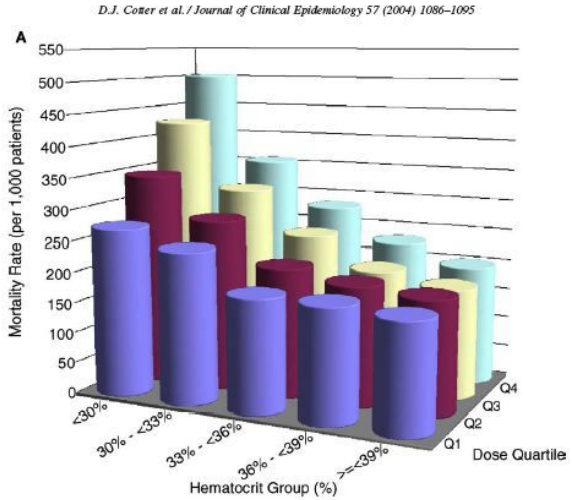


Figure 1. SRQ Plots of T_i/T_n (Vertical Axes) Against i/n (Horizontal Axes) for the Gibbs Sampler (a) and an Alternating Gibbs/Independence Sampler (b) for the Pump Failure Data Based on Runs of Length 5,000. Lines through the origin with unit slope are shown dashed; axis ranges are from 0 to 1 for all axes.



Mykland P, Tierney L, Yu B (1995) Regeneration in Markov chain samplers. Journal of the American Statistical Association 90:233-241, Figure 1, https://www.biostat.wisc.edu/~kbroman/topten_worstgraphs/

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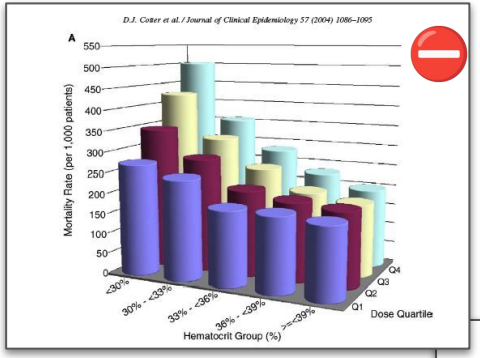
https://www.biostat.wisc.edu/~kbroman/topten_worstgraphs/cotter_fig2.html

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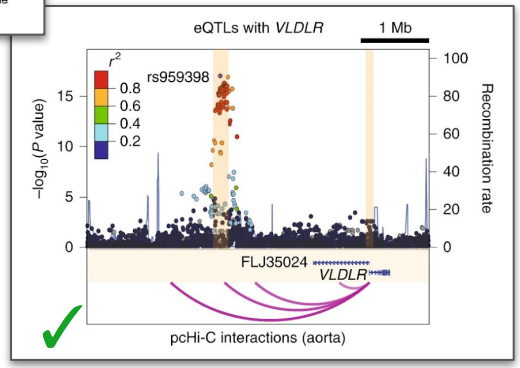
(4) Use simple and established ways to plot data

If readers don't need to understand the plot first, they can focus on the message:

- Use simple and established ways to visualise your data
  - Barplot, boxplot, scatterplot,...
- basic plots > specialised plots > first time visualisation
- Don't use pie charts or anything 3D



(use line chart instead)



(established, specialised figure)

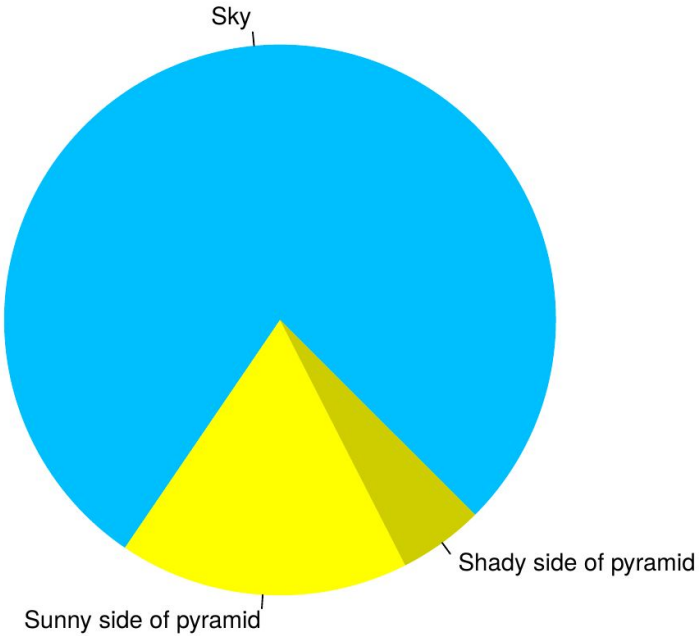
Jung, Inkyung, et al. "A compendium of promoter-centered long-range chromatin interactions in the human genome." Nature genetics 51.10 (2019): 1442-1449.

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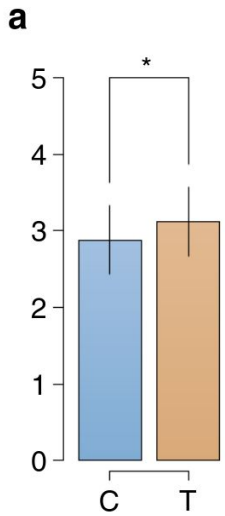
There are always exceptions

R

>?pie



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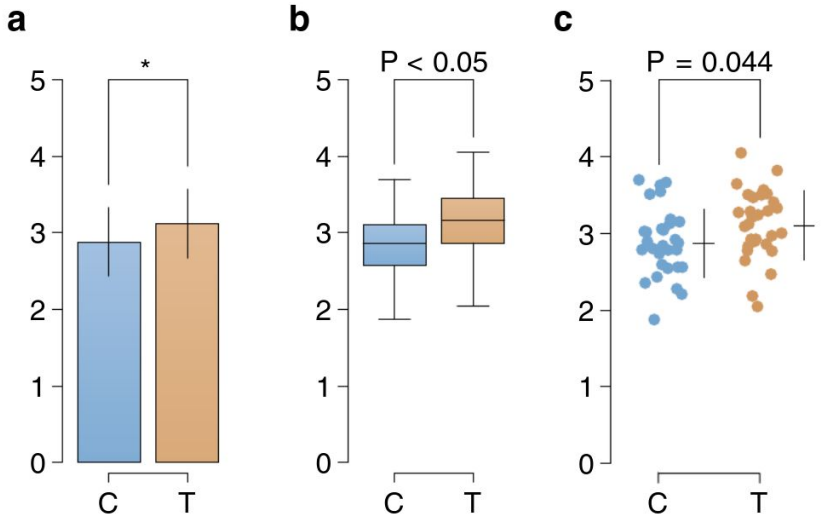


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(5) Original data vs summarised data

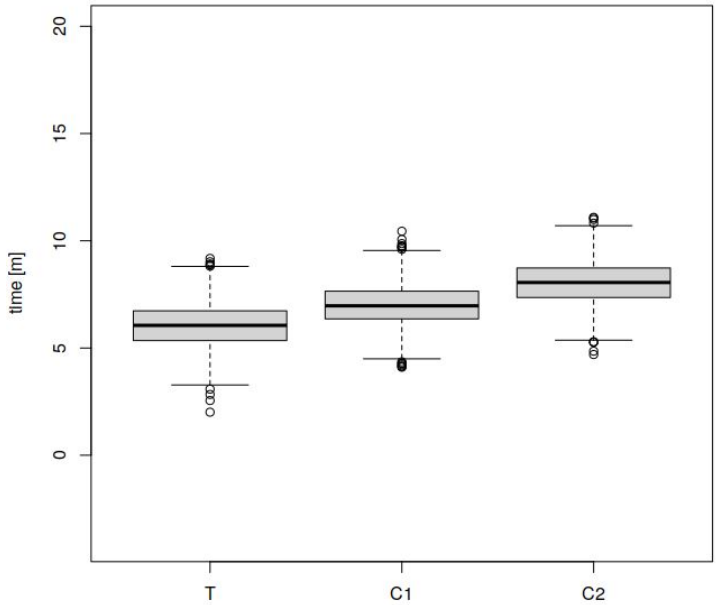
- Each figure interpretes data
- Showing the original data allows readers to interpret the data directly
- Increases confidence in the message (see (2))

-> show the original data (if possible and useful, see (1 and 4))



Ho, Joses, et al. "Moving beyond P values: data analysis with estimation graphics." Nature Methods 16.7 (2019): 565-566.
Streit, Marc, and Nils Gehlenborg. "Bar charts and box plots: creating a simple yet effective plot requires an understanding of data and tasks." Nature methods 11.2 (2014): 117-118.

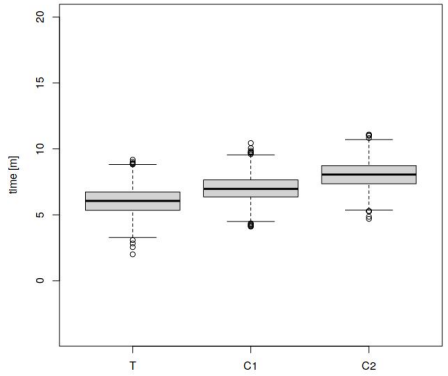
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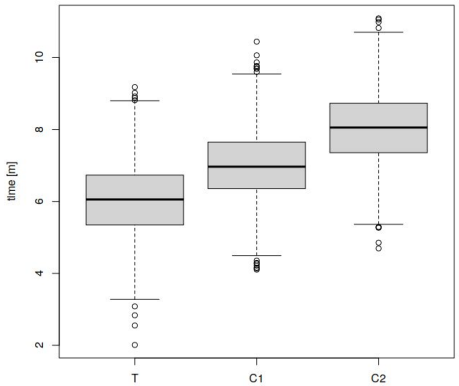


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(6) Minimise white space

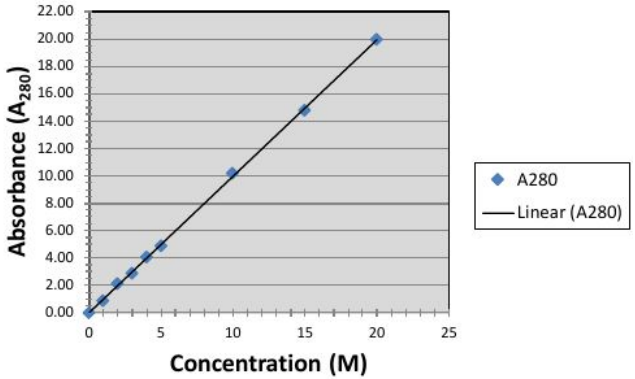
- Most of the space of your figure should be used to convey your message
- Minimise white space within figures
  - but use white space as “spacer” between figures, see multi-panel figures





https://genomicsclass.github.io/book/pages/plots_to_avoid.html

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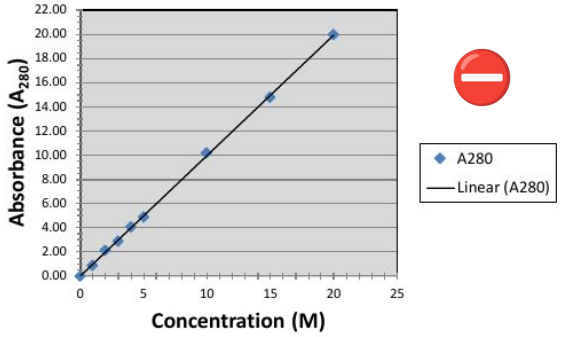


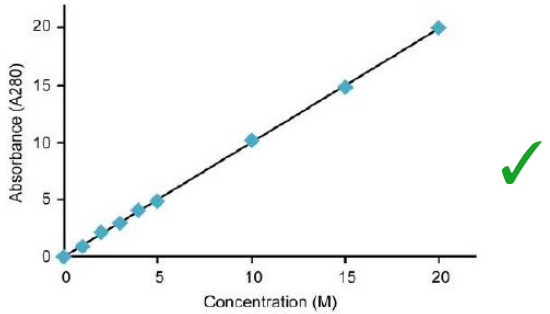
CREATING EFFECTIVE SCIENTIFIC FIGURES FOR PUBLICATION - AJE (2019)

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(7) Remove anything unnecessary

- The best figure shows the intended message at first glance
- Remove anything that is unnecessary to emphasise the content
  - Minimise tick marks, grid lines, duplicated legends, unnecessary colours, boxes, …
  - Avoid unwanted distortions of text, circles,



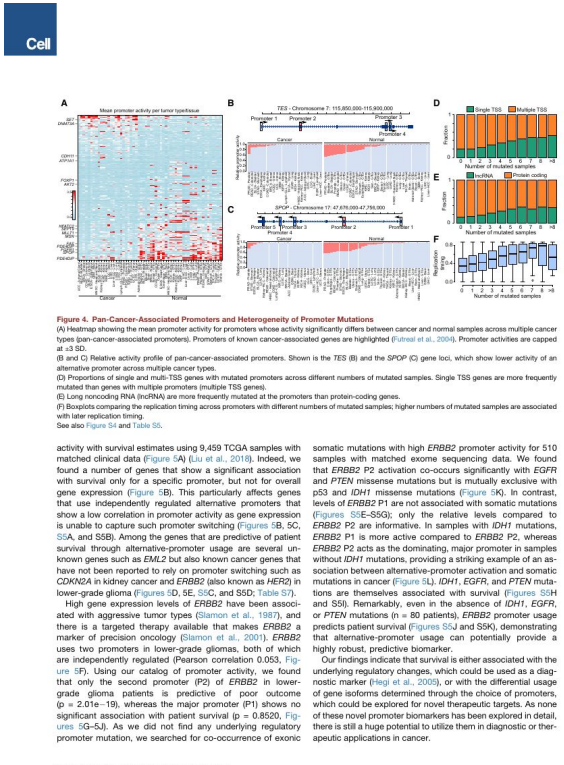


CREATING EFFECTIVE SCIENTIFIC FIGURES FOR PUBLICATION - AJE (2019)

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Rules for multi-panel figures

- Scientific figures are grouped together
- Each figure becomes a panel in a larger figure
- Many times this is not required, single panel figures can be more efficient for communication
  - Slides/Presentations
  - Discussion about results with manager
  - Markdown/Jupyter notebooks



Cell

Figure 4. Pan-Cancer-Associated Promoters and Heterogeneity of Promoter Mutations

(A) Heatmap showing the mean promoter activity for promoters whose activity significantly differs between cancer and normal samples across multiple cancer types (pan-cancer-associated promoters). Promoters of known cancer-associated genes are highlighted (Futreal et al., 2004). Promoter activities are capped at ±3 SD.

(B and C) Relative activity profile of pan-cancer-associated promoters. Shown is the *TES* (B) and the *SPOP* (C) gene loci, which show lower activity of an alternative promoter across multiple cancer types.

(D) Proportions of single and multi-TSS genes with mutated promoters across different numbers of mutated samples. Single TSS genes are more frequently mutated than genes with multiple promoters (multiple TSS genes).

(E) Long noncoding RNA (lncRNA) are more frequently mutated at the promoters than protein-coding genes.

(F) Boxplots comparing the replication timing across promoters with different numbers of mutated samples; higher numbers of mutated samples are associated with later replication timing.

See also Figure S4 and Table S5.

activity with survival estimates using 9,459 TCGA samples with matched clinical data (Figure 5A) (Liu et al., 2018). Indeed, we found a number of genes that show a significant association with survival only for a specific promoter, but not for overall gene expression (Figure 5B). This particularly affects genes that use independently regulated alternative promoters that show a low correlation in promoter activity as gene expression is unable to capture such promoter switching (Figures 5B, 5C, S5A, and S5B). Among the genes that are predictive of patient survival through alternative-promoter usage are several unknown genes such as *EML2* but also known cancer genes that have not been reported to rely on promoter switching such as *CDKN2A* in kidney cancer and *ERBB2* (also known as *HER2*) in lower-grade glioma (Figures 5D, 5E, S5C, and S5D; Table S7).

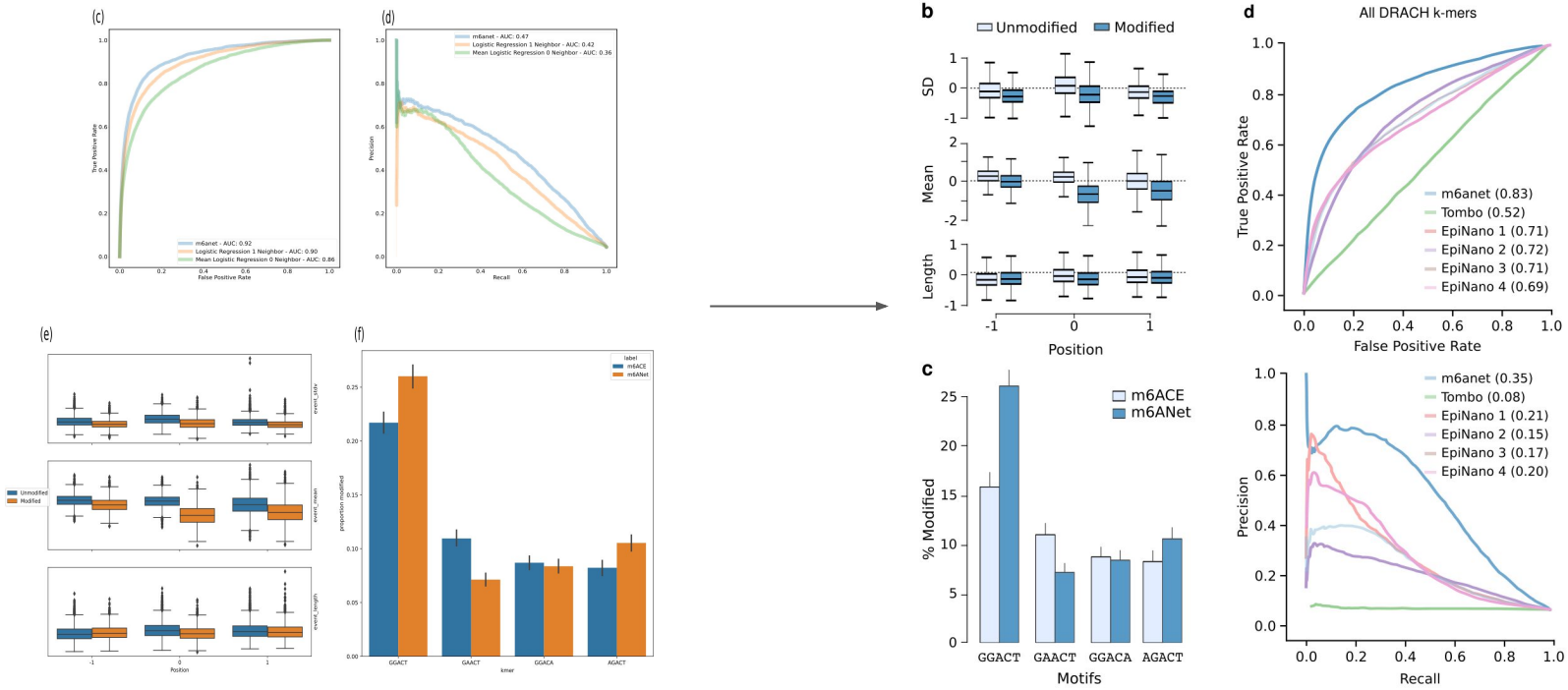
High gene expression levels of *ERBB2* have been associated with aggressive tumor types (Slamon et al., 1987), and there is a targeted therapy available that makes *ERBB2* a marker of precision oncology (Slamon et al., 2001). *ERBB2* uses two promoters in lower-grade gliomas, both of which are independently regulated (Pearson correlation 0.053, Figure 5F). Using our catalog of promoter activity, we found that only the second promoter (P2) of *ERBB2* in lower-grade glioma patients is predictive of poor outcome ($p = 2.01e{-}19$), whereas the major promoter (P1) shows no significant association with patient survival ($p = 0.8520$, Figures 5G–5J). As we did not find any underlying regulatory promoter mutation, we searched for co-occurrence of exonic somatic mutations with high *ERBB2* promoter activity for 510 samples with matched exome sequencing data. We found that *ERBB2* P2 activation co-occurs significantly with *EGFR* and *PTEN* missense mutations but is mutually exclusive with p53 and *IDH1* missense mutations (Figure 5K). In contrast, levels of *ERBB2* P1 are not associated with somatic mutations (Figures S5E–S5G); only the relative levels compared to *ERBB2* P2 are informative. In samples with *IDH1* mutations, *ERBB2* P1 is more active compared to *ERBB2* P2, whereas *ERBB2* P2 acts as the dominating, major promoter in samples without *IDH1* mutations, providing a striking example of an association between alternative-promoter activation and somatic mutations in cancer (Figure 5L). *IDH1*, *EGFR*, and *PTEN* mutations are themselves associated with survival (Figures S5H and S5I). Remarkably, even in the absence of *IDH1*, *EGFR*, or *PTEN* mutations ($n = 80$ patients), *ERBB2* promoter usage predicts patient survival (Figures S5J and S5K), demonstrating that alternative-promoter usage can potentially provide a highly robust, predictive biomarker.

Our findings indicate that survival is either associated with the underlying regulatory changes, which could be used as a diagnostic marker (Hegi et al., 2005), or with the differential usage of gene isoforms determined through the choice of promoters, which could be explored for novel therapeutic targets. As none of these novel promoter biomarkers has been explored in detail, there is still a huge potential to utilize them in diagnostic or therapeutic applications in cancer.

Demircioglu et al (2019)

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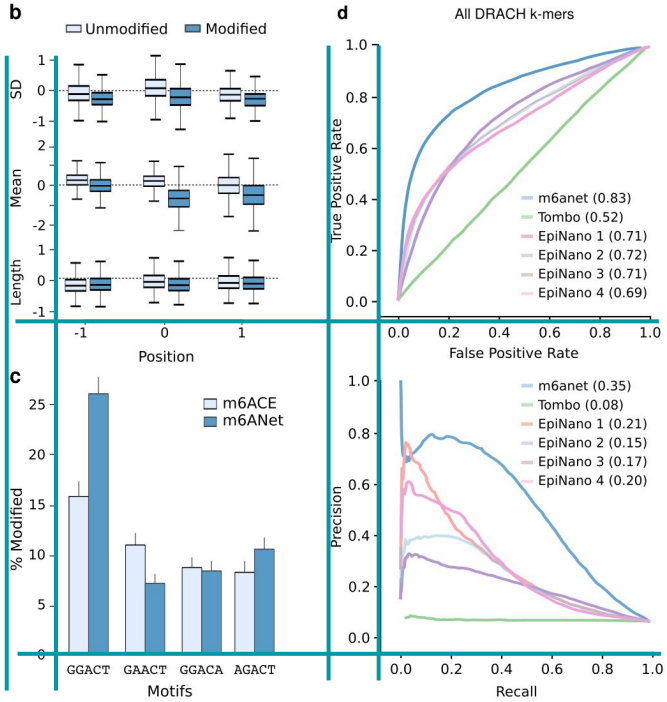
Multi-panel figures



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(8) Align everything (invisible lines/grid)

- Every alignment creates a visual line
  - Text, labels, axes, legends, …
- These visual lines form a grid
- All visual lines should be aligned such that the figure uses a clear grid

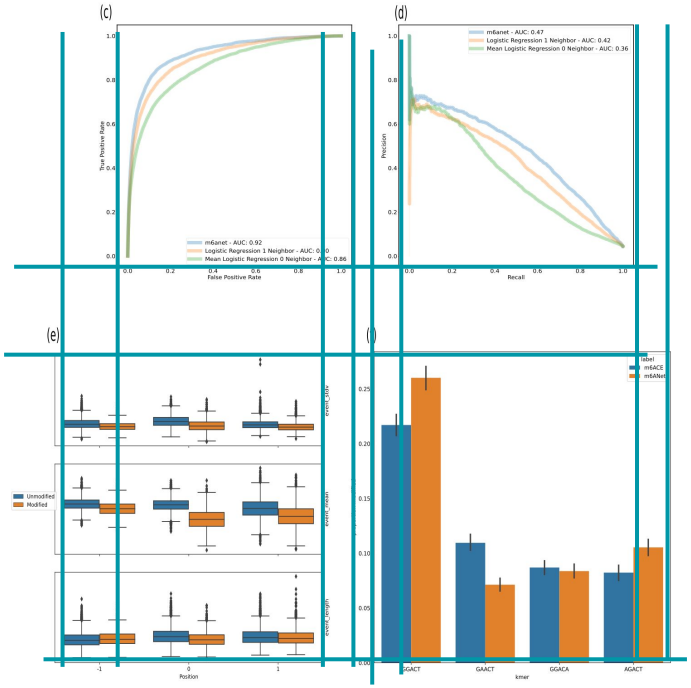


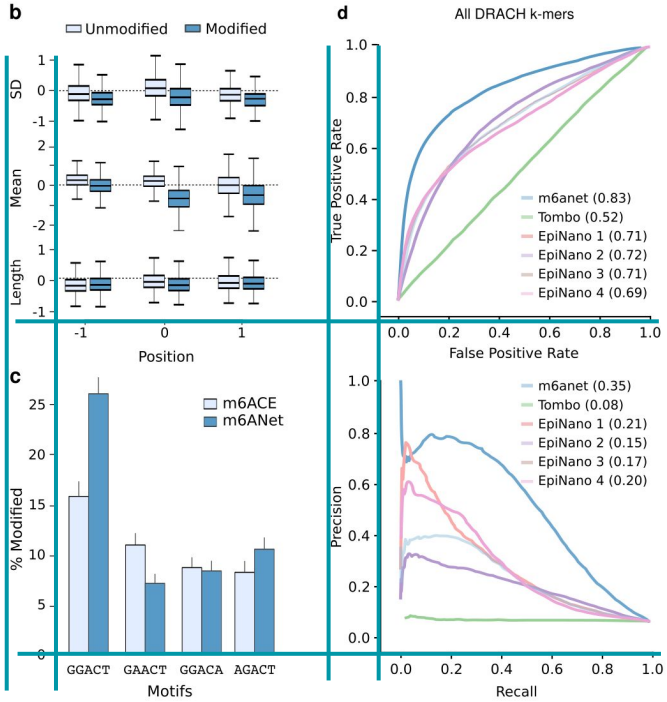
✓

https://bioinformatics-core-shared-training.github.io/effective-figure-design/DesigningEffectiveScientificFigures_Zabala_afternoon_v00.pdf

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Multi-panel figures

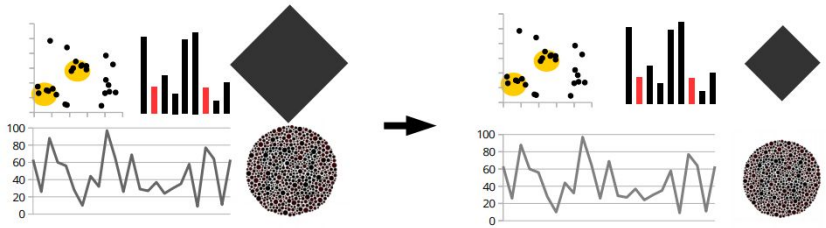


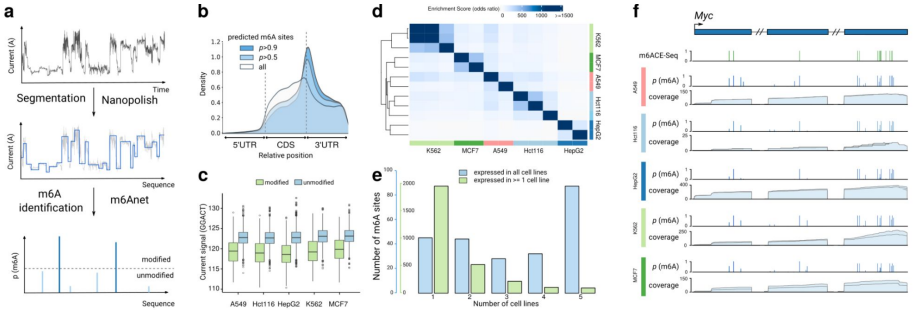


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(9) Balance the plot to emphasise your messages

- Colour and space can change the emphasis of elements in a figure
- Visual weight should be balanced (avoid saturated colours)
- Figures space should be balanced





https://bioinformatics-core-shared-training.github.io/effective-figure-design/DesigningEffectiveScientificFigures_Zabala_afternoon_v00.pdf

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(10) Be consistent across panels and figures

- Use the same font size for all figures
- Use the same colour scheme for all figures
  - saturated colours are better avoided
- Use the same plots/metrics for the same data

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Figure Design: Software

- High quality figures are designed using vector graphics
- Save figures as svg/pdf
- AI tools can help with generating figures and designing figures
  - Always review figures and the underlying results!
- Final edits : Inkscape (or Adobe Illustrator)



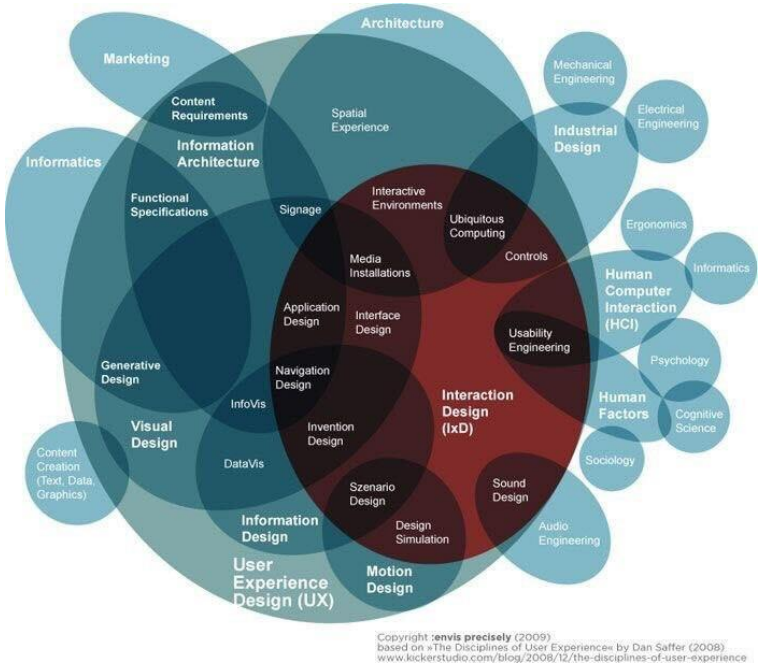


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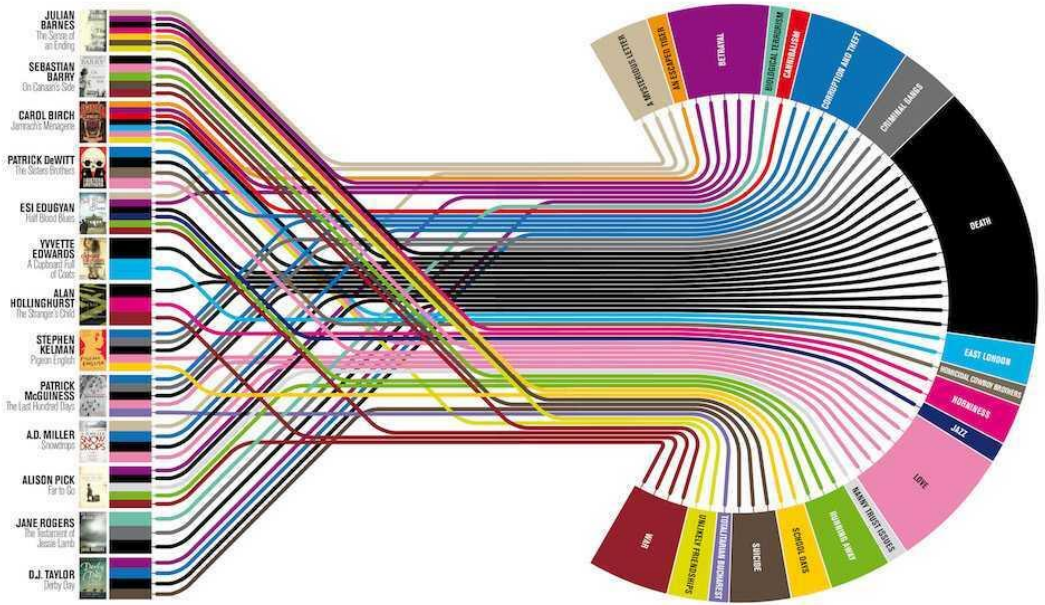
Summary: 10 Rules for better figures

- Define the message
- Select the simplest and best known visualisation to show that message
- Show raw data if possible
- Remove any distraction from message
  - Be consistent (font, plot types, colors,...)
  - Minimise white space
  - Align panels
  - Balance figures

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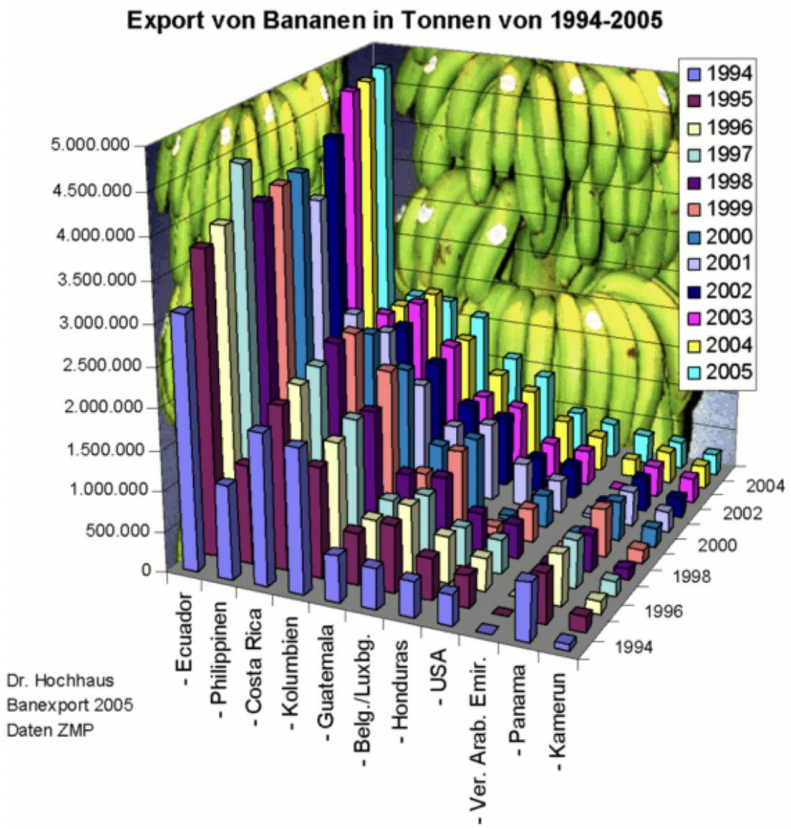


Plot lines

What makes a prize-winning novel? As Julian Barnes wins the Booker Prize, Delayed Gratification's Johanna Kamradt charts the themes of this year's longlisters.

Illustration: Christian Tate

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Links

- Ten Simple Rules for Better Figures, Nicolas P. Rougier, Michael Droettboom, Philip E. Bourne
  - https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1003833
- https://www.aje.com/dist/docs/Guide-Creating-Effective-Scientific-Figures-for-Publication.pdf
- https://bioinformatics-core-shared-training.github.io/effective-figure-design/DesigningEffectiveScientificFigures_Zabala_afternoon_v00.pdf
- http://blogs.nature.com/methagora/2013/07/data-visualization-points-of-view.html
- Moving beyond P values: data analysis with estimation graphics: https://www.nature.com/articles/s41592-019-0470-3.epdf?author_access_token=Euy6APITxsYA3huBKOFBvNRgN0jAjWel9jnR3ZoTv0Pr6zJiJ3AA5aH4989gOJS_dajtNr1Wt17D0fh-t4GFcvqwMYN03qb8C33na_UrCUcGrt-Z0J9aPL6TPSbOxIC-pbHWKUDo2XsUOr3hQmlRew%3D%3D

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Analysing m6A in human cell lines

(Task 2)

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Task 2: Prediction of m6A sites in all SG-NEx direct RNA-Seq data sets

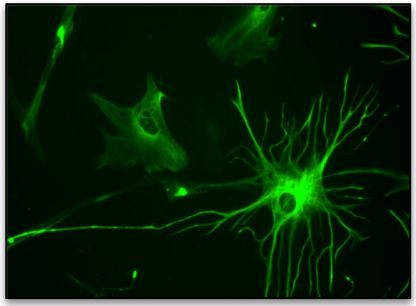
- **Predict m6A RNA modifications** in all direct RNA-Seq data sets from the **SG-NEx data** using your method
- **Describe the results** and **compare them** across the different cell lines
- **Summarise** and **visualise** your observations
- Create an **interactive data visualisation platform** for other researchers to explore this data

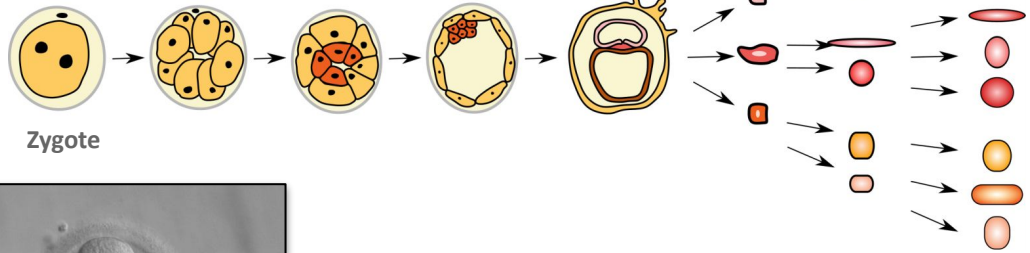
The task is broadly defined on purpose:

- Demonstrate that you can interpret data and predictions within a specific domain/context
- Identify relevant questions
- Communicate results

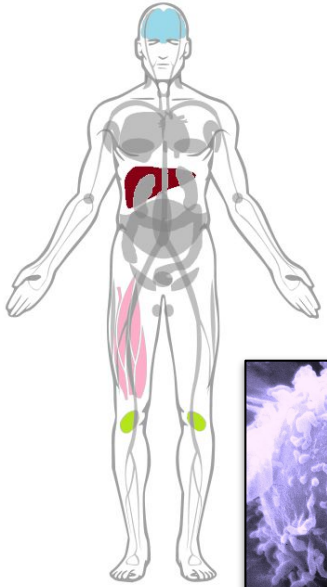
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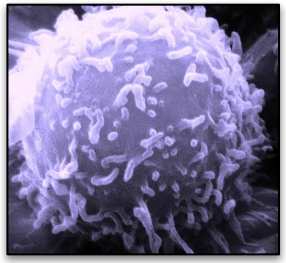
Genes, Cells, Tissues, Organs: the Human Body









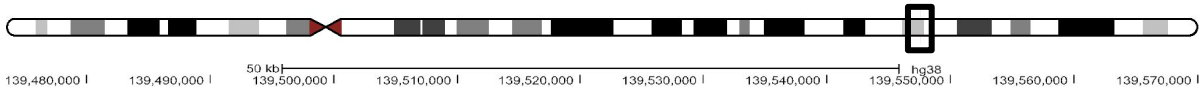


>200 cell types, >35 trillion cells

Image: http://www.ebi.ac.uk Expression Atlas

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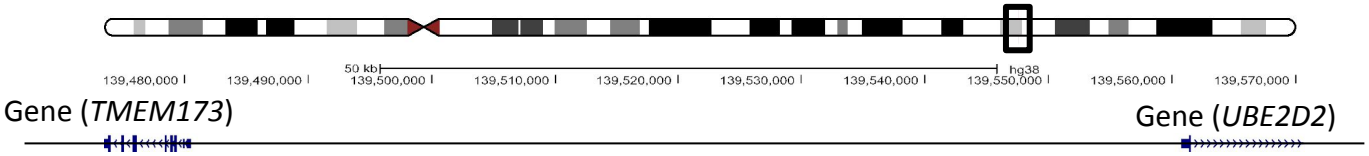
Genes and Gene Expression



- **Gene expression**: Process in which a gene is activated and a functional gene product (RNA or protein) is synthesized

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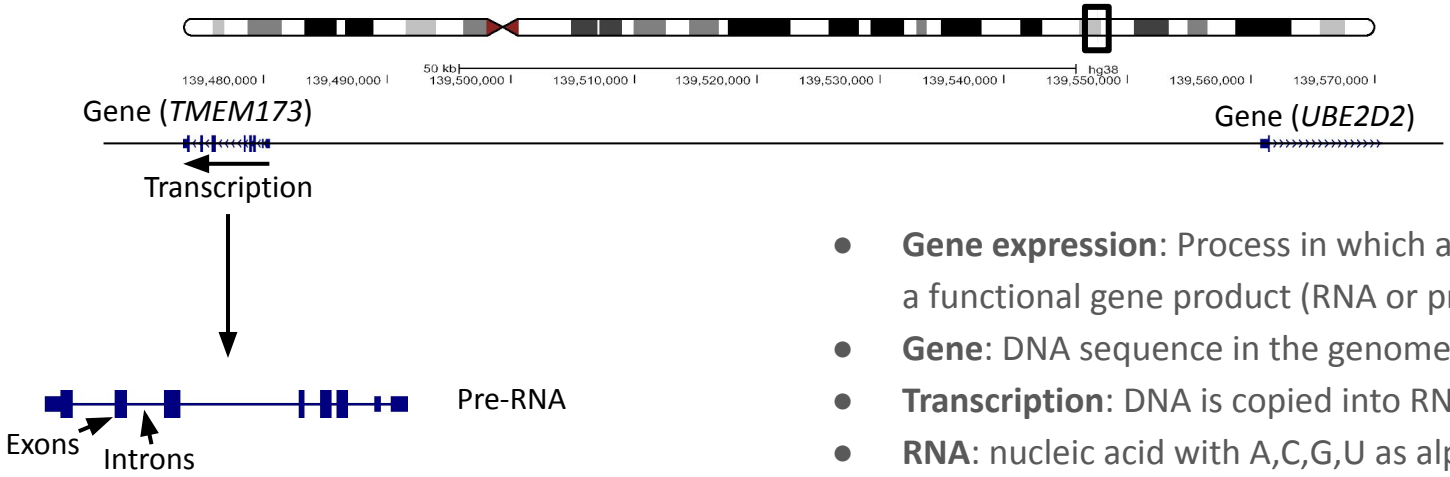
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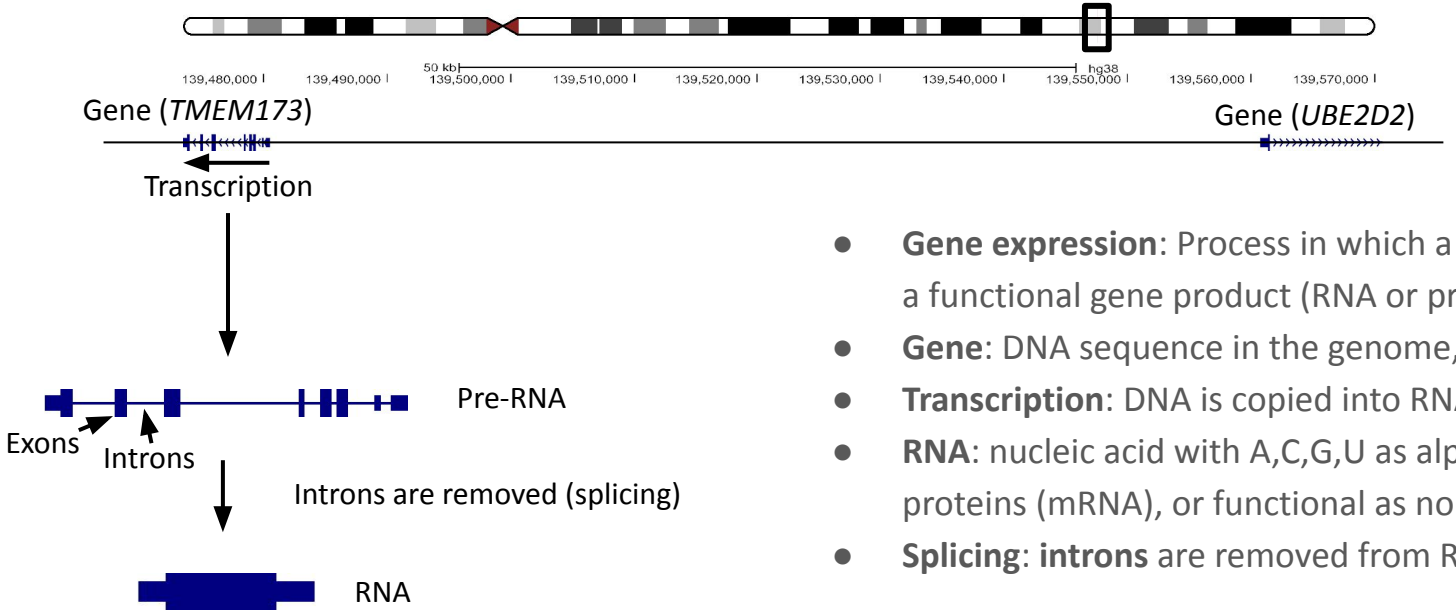
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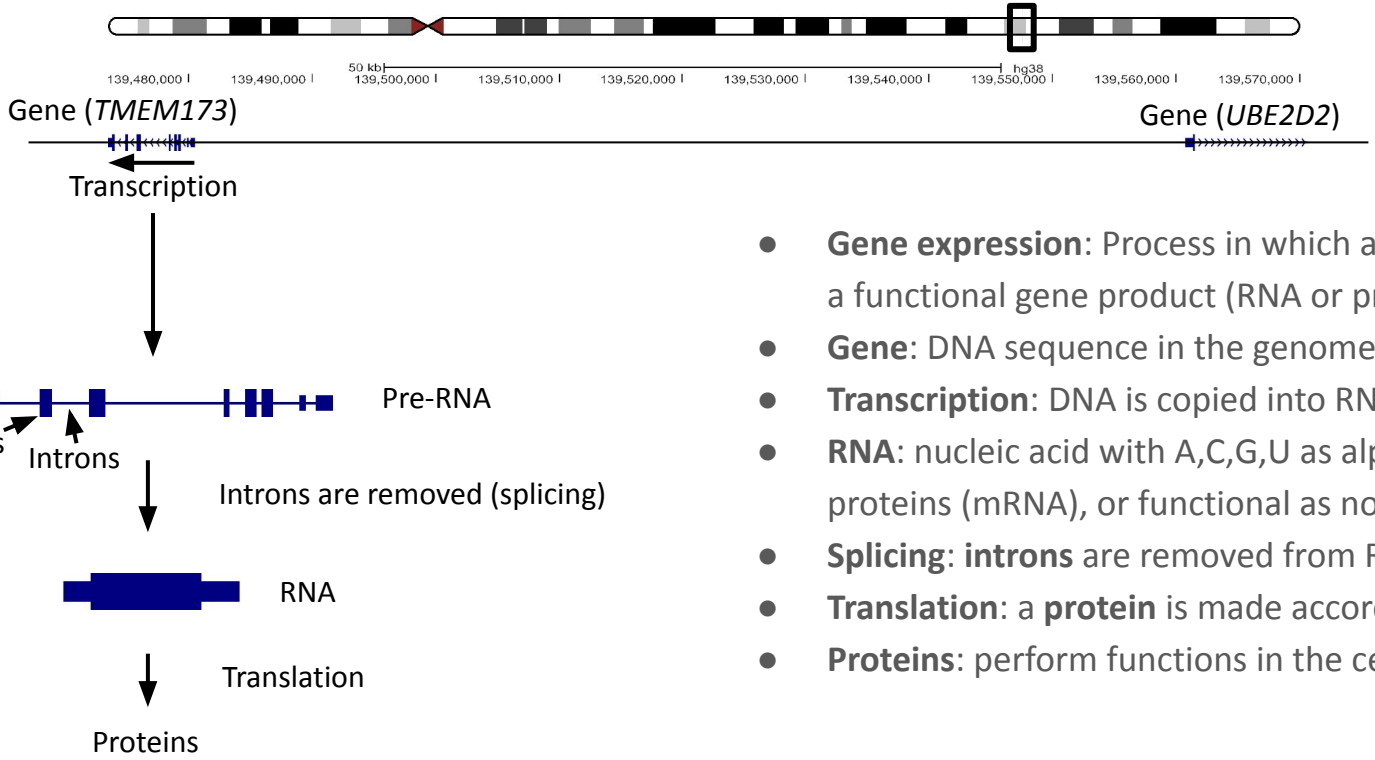
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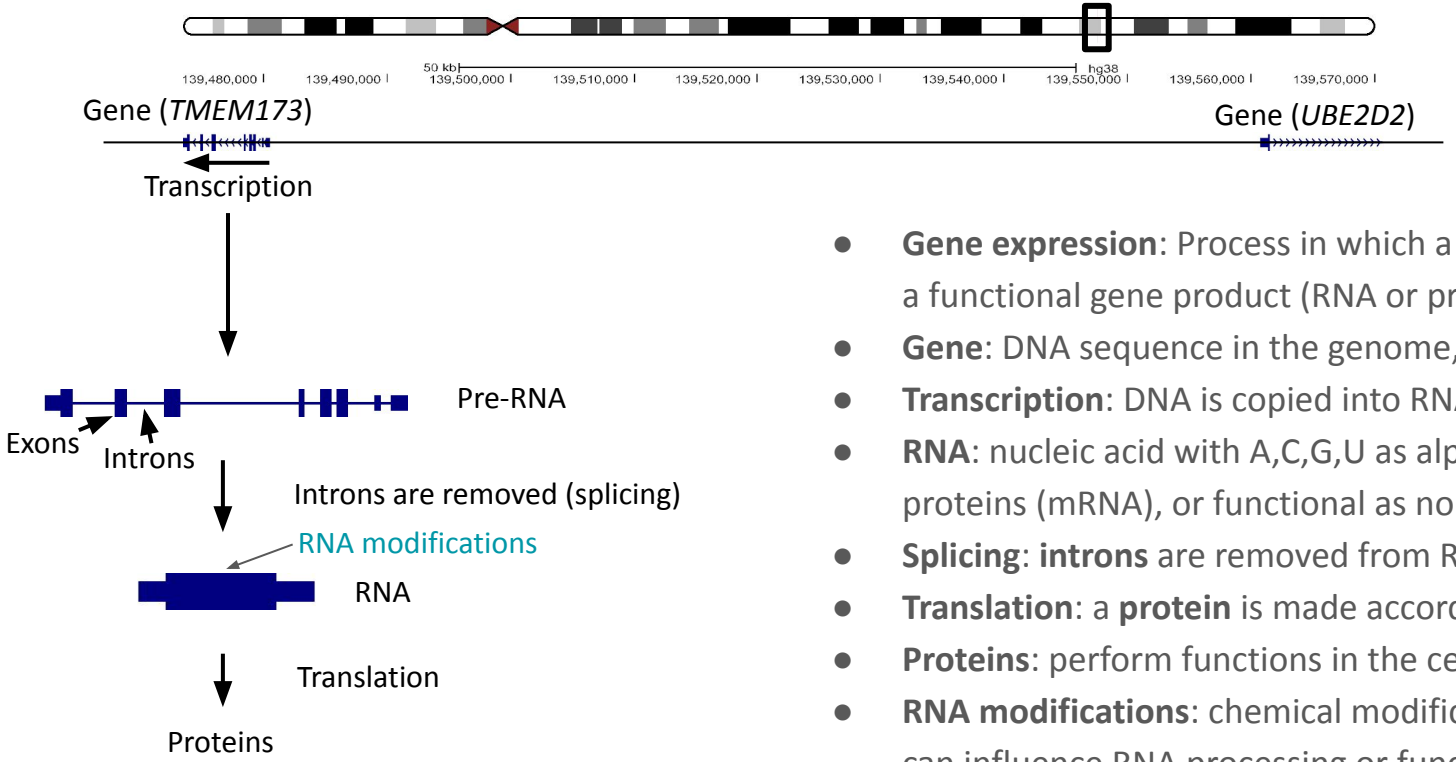
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- **Proteins**: perform functions in the cell

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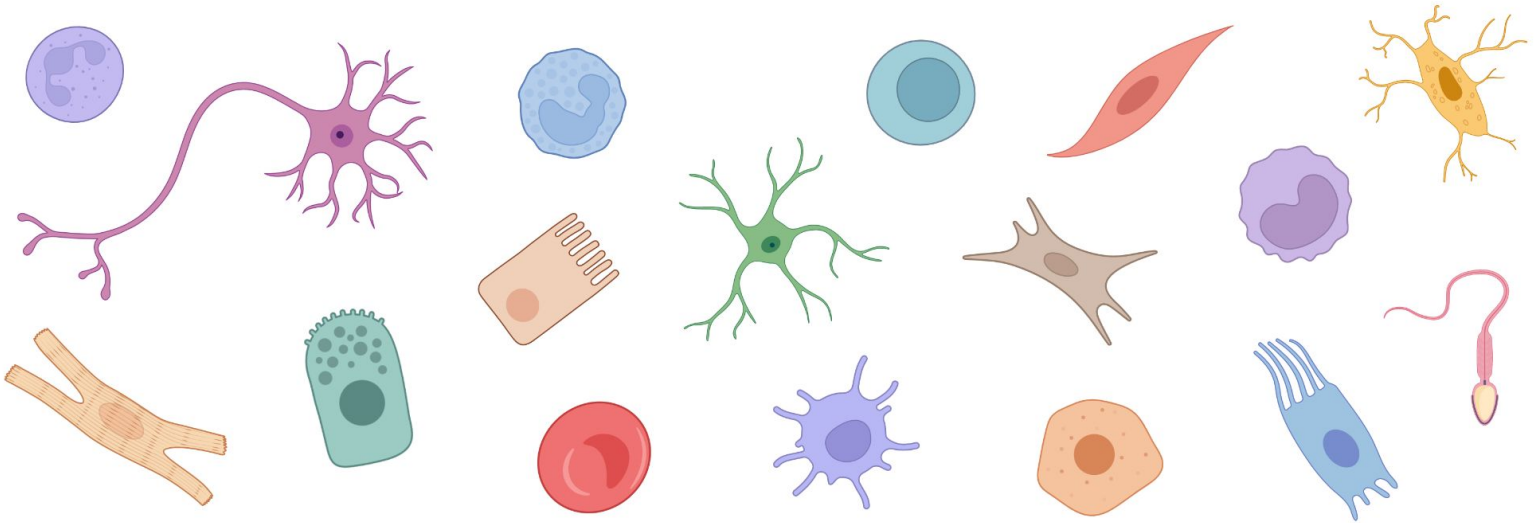
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- **Translation**: a **protein** is made according to RNA
- **Proteins**: perform functions in the cell
- **RNA modifications**: chemical modifications of RNA that can influence RNA processing or function

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The Genome, Transcriptome, and Epitranscriptome



Only some genes are active at any time:

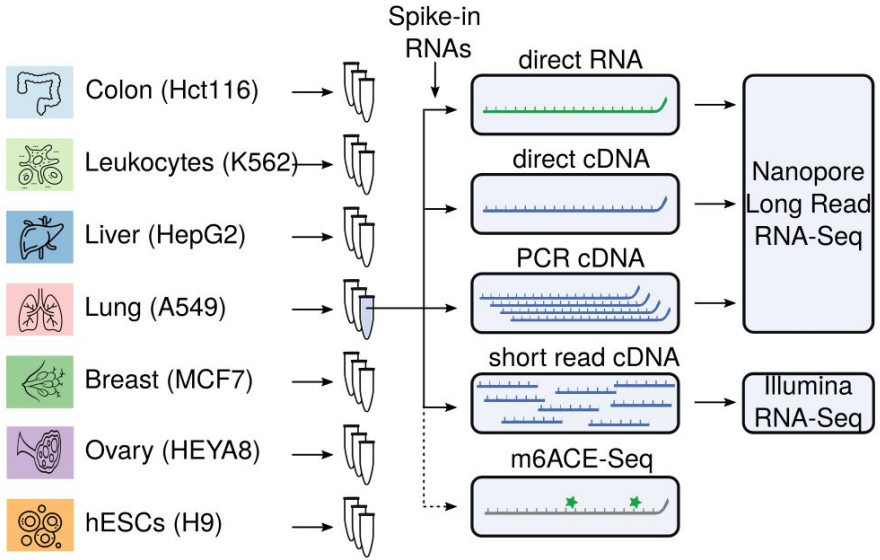
One genome -> many different cells

What about m6A modifications?

Illustration: Biorender

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The SG-NEx Project



aws open data

https://github.com/GoekeLab/sg-nex-data

Chen et al (Nature Methods, 2025): https://www.nature.com/articles/s41592-025-02623-4

National Cancer Centre Singapore SingHealth

NUS National University of Singapore

Peter Mac Peter MacCallum Cancer Foundation

NUH National University Hospital

The University of North Carolina at Chapel Hill

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The SG-NEx data on AWS

- The direct RNA-Seq data was processed with m6Anet to generate the data.json files.
- The SG-NEx data can be accessed through AWS
  - the S3 bucket can be browsed with this link: https://sg-nex-data.s3.amazonaws.com/index.html
  - the m6Anet processed files can be found under *data/processed_data/m6Anet/*
- The data access tutorial shows how to access the data using the AWS command line interface
  - https://github.com/GoekeLab/sg-nex-data/blob/master/docs/AWS_data_access_tutorial.md#processed-data

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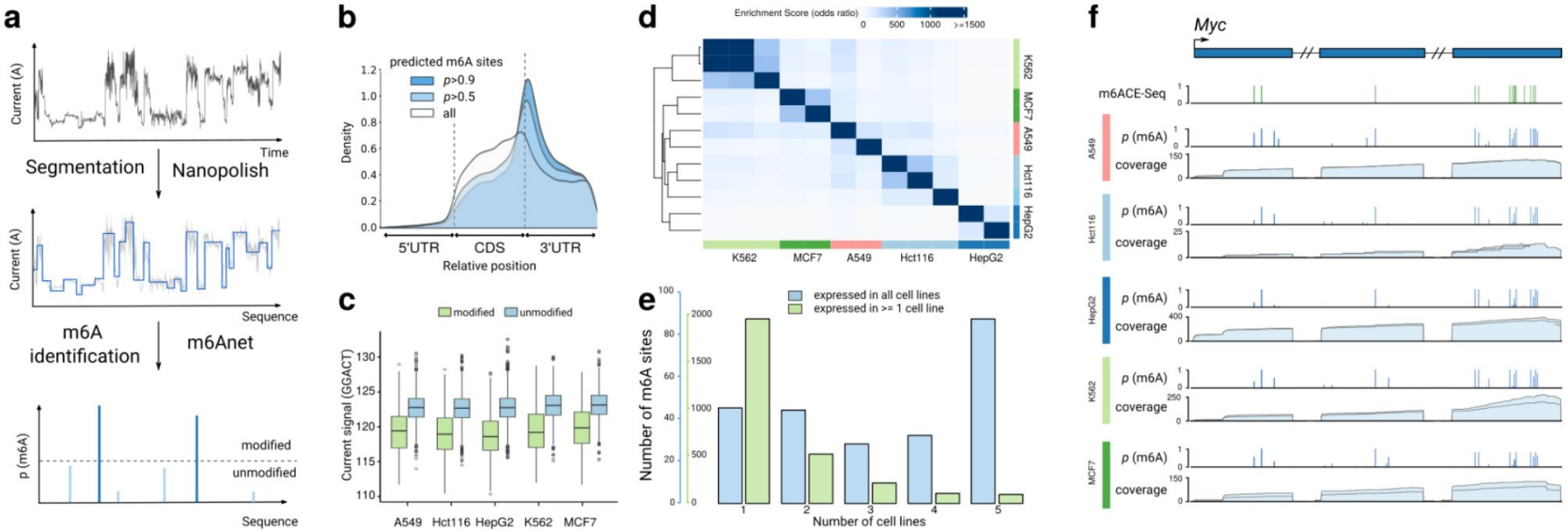
m6A: Processed files from the SG-NEx data set



- Direct RNA-Seq from 7 cell lines is available
- Each cell line has several replicate experiments
- These files have the same format as the data.json files, you can analyse them using your method

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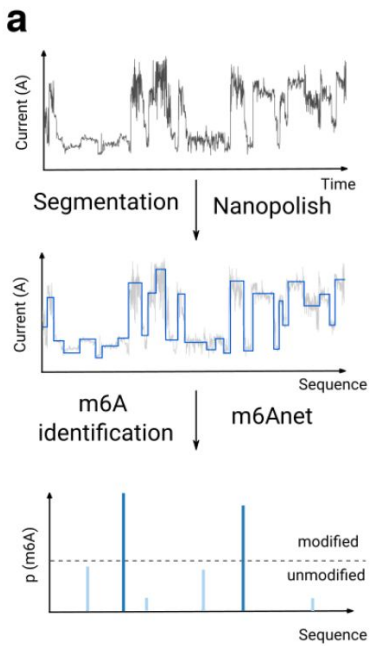
Profiling of m6A in 5 cell lines using direct RNA-Seq



Chen et al (Nature Methods, 2025): https://www.nature.com/articles/s41592-025-02623-4

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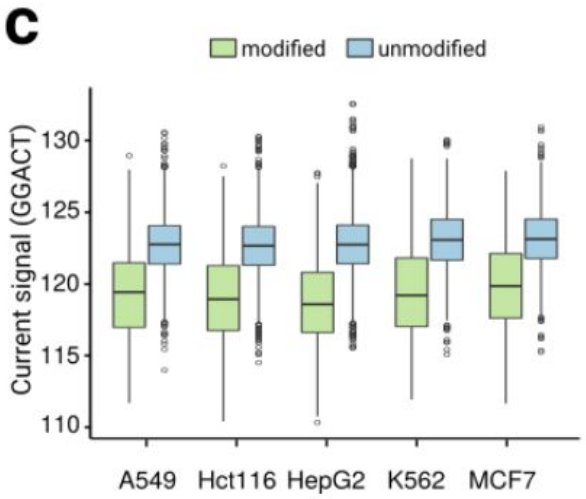
Illustrate the approach



- Shows the data processing and analysis approach
- Introduces visualisation of results

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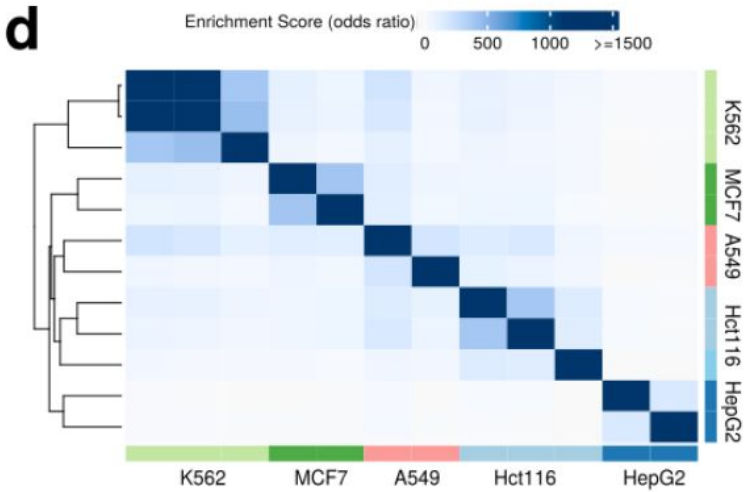
Analyse the features that you are using



- Which features differ between modified and unmodified sites? Does this look similar across the cell lines?

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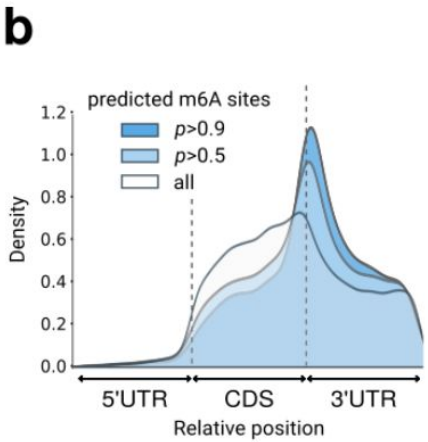
Compare m6A across all cell lines



- Do you observe that m6a is similar between cell lines, or different? Which cell lines are more similar? How many sites are shared, how many are unique?

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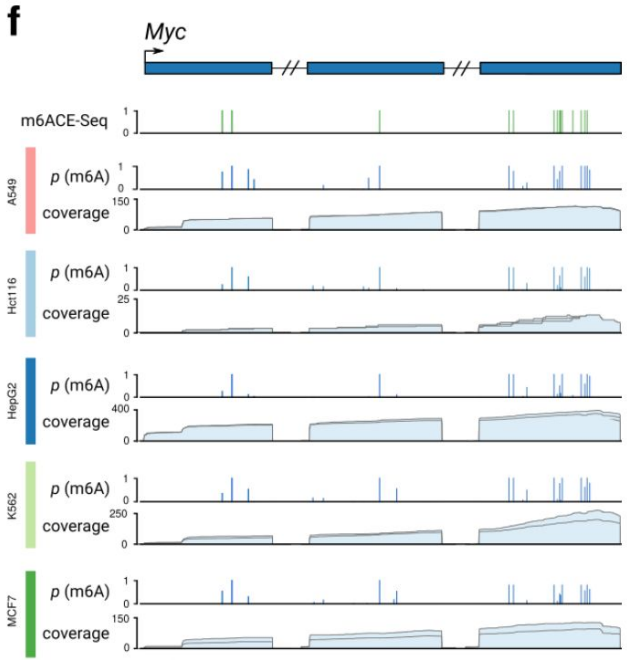
Relative location of m6A along mRNAs



- m6A is expected to occur between the coding sequence (CDS) and the 3’ untranslated region (UTR)
- The metagene plot shows the relative position of detected m6A sites across all transcripts

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Investigate individual sites/genes



- Which genes are highly modified?
- Which genes show differences across cell lines?
- Visualisation of individual gene examples can illustrate results

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Interactive data visualisation platform

- Create an interactive data visualisation platform for other researchers to explore this data
- There are many different aspects of the project that can be visualised
- Teams are welcome to define the scope themselves
- You can describe the purpose and target audience

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Task 2: Prediction of m6A sites in all SG-NEx direct RNA-Seq samples

Tasks:

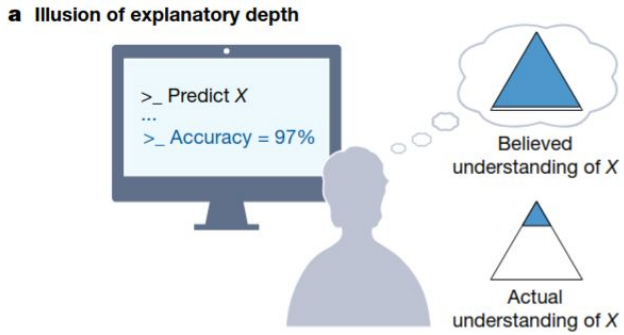
- Predict m6A RNA modifications in all direct RNA-Seq data sets from the SG-NEx data using your method
- Describe the results and compare them across the different cell lines
- Summarise and visualise your observations
- Create an interactive data visualisation platform for other researchers to explore this data

Advice and things to consider:

- Present an overview of the data, summarise the data
- Think about possible questions/hypothesis
- How can you summarise and visualise your results?

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Tips for working with AI



Artificial intelligence and illusions of understanding in scientific research. Messeri & Crockett, 2024, Nature.

AI reflection: What professional strategies could you employ in future work?

"Consulting an expert" is often not an option -> how can you make yourself the expert?

AI risk #1: Generating a plausible but incorrect answer

- Inspect the data in different ways
- Are there different ways you can answer the question? Do they give the same answer?

AI risk #2: Not fully understanding the context or data you're working with

- Do you understand all the datasets & references you're using?
- Where did they come from? How were they generated?

AI risk #3: Not fully thinking about the question or problem

- Are you actually answering the problem at hand?

AI risk #4: Overconfidence in understanding

- Has the AI output given you a false sense of confidence?
- Can you ask AI to test you? Ask you to explain concepts you should understand?
- Could you explain it simply? (If not - you might not fully understand it)

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Upcoming deadlines

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Upcoming deadlines

- Team Project submission: **Oct 28th 2026 (Wednesday), 23.59pm**
  - Final leaderboard submission
  - Report
  - Visualisation
  - Code repository
- Individual Submission: **Oct 29th 2026 (Thursday), 23.59pm**
  - Teamwork Summary
- Student Evaluation of Team Projects: **Nov 15th (Sunday), 23.59 pm**
  - Peer evaluation of the code documentation and code reproducibility

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Hackathon 3/4

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Hackathon 3 of 4

Please prepare & post on the team group zulip channel:

1. An overview of the current project status to discuss (a few slides can help!)
2. The specific aims for this session, and who will be working on this

After the hackathon (by 23:59) please post on our group zulip channel

1. What was achieved
2. What will be done by each team member until the next hackathon



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Meeting the teams - Schedule

19/10/26

Clare	Big Study Group
Clare	datahelix
Clare	Dolly
Clare	foursight

Jonathan	Genomies
Jonathan	Genomix/haribo
Jonathan	Lifo
Jonathan	Sonic_hedgehog
Jonathan	yellow

26/10/25

Clare	Genomies
Clare	Genomix/haribo
Clare	Lifo
Clare	Sonic_hedgehog
Clare	yellow

Jonathan	Big Study Group
Jonathan	datahelix
Jonathan	Dolly
Jonathan	foursight

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